Lab: Long-Term Innovation Strategy Design

Objective

Design a long-term innovation strategy for your invention that integrates technology roadmapping, portfolio thinking, exploration-exploitation management, and adaptive planning. You will create a multi-horizon planning document that is simultaneously ambitious and responsive to evidence.

Materials and Prerequisites

- Your invention concept and all analyses from Modules 01-07
- Understanding of Active Inference planning from Section 1

Part 1: Technology Roadmap Construction (15 minutes)

Step 1: State your invention's long-term vision (5-10 year horizon). What does the world look like if your invention succeeds at scale?

Write your response here

Step 2: Work backward from the vision to define three milestones (roughly at 1-year, 3-year, and 5-year horizons).

Milestone	Target Date	Description	Key Achievement	Uncertainty Level (1-10)
M1 (Near-term)				
M2 (Mid-term)				
M3 (Long-term)				

Step 3: For each milestone, specify decision criteria and revision conditions.

Milestone	Evidence of On-Track	Evidence of Off-Track	Revision Plan If Off-Track
M1			
M2			
M3			

Step 4: Identify the "monkey" — the hardest, most uncertain element of your roadmap. What is the one thing that, if it proves impossible, makes the rest of the roadmap irrelevant?

Write your response here

Step 5: How would you test the "monkey" before investing heavily in the rest of the roadmap?

Write your response here

Part 2: Innovation Portfolio Design (10 minutes)

Even if you are focused on a single invention, portfolio thinking helps you allocate attention and resources across different types of activities.

Step 1: Classify your current and planned activities across the three innovation horizons.

Horizon	Activities	% of Resources	Risk Level	Expected Timeline to Returns
H1: Core optimization				
H2: Adjacent expansion				
H3: Transformative innovation				

Step 2: Evaluate your portfolio balance. Are you over-invested in any horizon? Under-invested?

Write your response here

Step 3: If you could add one activity to your portfolio, what would it be and which horizon would it serve?

Write your response here

Part 3: Exploration-Exploitation Balance (10 minutes)

Step 1: Assess your current balance between exploration and exploitation.

Dimension	Exploration Activities	Exploitation Activities	Current Balance
Technology			
Market			
Business model			
Team/capabilities			

Step 2: Is your environment (market, technology, regulation) currently stable or rapidly changing? How does this affect the optimal exploration-exploitation balance?

Write your response here

Step 3: Design one structural mechanism to protect exploration from the gravitational pull of exploitation. How would this mechanism work in practice?

Write your response here

Part 4: Innovation Pipeline Design (15 minutes)

Design a pipeline that moves ideas from concept to market with appropriate evidence requirements at each stage.

Step 1: Define four stages for your innovation pipeline, with gate criteria that match the stage's uncertainty level.

Stage	Purpose	Duration	Investment Level	Gate Criteria (What evidence is required to proceed?)
Stage 1: Discovery				
Stage 2: Validation				
Stage 3: Development				
Stage 4: Launch				

Step 2: For each gate, specify a "kill criterion" — evidence that would cause you to terminate the project rather than proceed.

Gate	Kill Criterion	Why This Evidence Is Decisive
Gate 1		
Gate 2		
Gate 3		

Step 3: How would you ensure that early-stage gates are not too demanding? What mechanisms prevent premature killing of ideas that have not yet had a chance to generate evidence?

Write your response here

Part 5: Adaptive Strategy Integration (10 minutes)

Step 1: Define your "commander's intent" — the strategic vision and principles that remain constant even as specific plans change.

Element	Your Definition
Strategic vision (end state)	
Core principle 1	
Core principle 2	
Core principle 3	
What is NOT negotiable	
What IS flexible	

Write your response here

Step 2: Design a strategic review process. How often will you formally evaluate whether your plans need revision? What triggers an emergency review?

Review Type	Frequency	Participants	Key Questions	Output
Routine review				
Milestone review				
Emergency review				

Step 3: What is the one strategic question you most need to answer in the next 90 days? How will you answer it?

Write your response here

Discussion and Debrief

1. How did the "monkey-first" principle change your roadmap? Were there elements of your plan that you had been avoiding because they were the hardest?

2. Was your innovation portfolio balanced, or did you find yourself over-weighted toward one horizon? What organizational or personal factors create this imbalance?

3. How difficult was it to design structural protections for exploration? What pressures in your environment push you toward exploitation?

4. Did the pipeline gate criteria exercise reveal any stages where you were demanding too much or too little evidence? How does the Active Inference principle of matching precision to stage clarify gate design?

5. How does the concept of "commander's intent" change your approach to planning? Is it easier or harder to plan with fixed principles and flexible tactics than with fixed plans?